Report on Neuro-Symbolic Artificial Intelligence: Frameworks for Multiresolution Social and Spiritual Reasoning

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Abstract

This document serves as a comprehensive system summary and architectural manifest detailing the design, implementation, and philosophical foundations of our Neuro-Symbolic AI framework. By unifying deep learning pattern-recognition matrices with traditional rule-based symbolic AI, the system realizes unprecedented out-of-distribution generalization, transferability, and explainability across diverse social, institutional, and spiritual narratives.

1 Introduction and Architectural Core

Neuro-Symbolic AI (NeSy) is defined as any system that combines the sub-symbolic pattern-recognition capabilities of deep learning (neural networks) with the logical, rule-based reasoning of traditional computer science (symbolic AI). Our research implements a multi-tier pipeline explicitly configured to satisfy five core desiderata: deep qualitative reasoning, robust out-of-distribution generalization, complete semantic interpretability, efficient sample learning from sparse data exemplars, and transactional domain transferability.

The unified system architecture is organized across four distinct operational layers:

- **Intent:** The teleological definition of narrative, spiritual, and multi-agent social structures.
- **Specifications:** The input-output interface mappings and vocabulary constraints bounding the environment.
- **Context:** The dynamic extraction of situational semantics via interlocking decision forest memories.
- **Training and Inferencing:** The optimization and execution plane where agents plan, reason, and learn under uncertainty.

2 The Intent Layer: Hierarchical Reasoning Taxonomy

The core intent of the system is to investigate and deploy formal methods for an AI framework capable of mentalizing and reasoning over highly non-convex social, institutional, and spiritual matters. Bible stories and contemporary situations are combined to train the underlying models. The cognitive training tracks are organized as a multiresolution document hierarchy:

2.1 Individual Reasoning (The Mind Domain)

- **Inductive Reasoning:** Generating generalized concepts from sparse spatial observations.
- **Logical Reasoning:** Formal deductive theorem proving over strict axiomatic bases.
- **Analogical Reasoning:** Mapping structural isomorphisms to facilitate the cross-domain transfer of knowledge.
- **Algebraic Reasoning:** Symbolic operations over discrete algebraic group structures.
- **Continuous Reasoning:** Continuous trajectory modeling within time-bounded lattices.
- **Problem Solving:** Joint mechanical, scientific, and analogical problem resolution.
- **Intensional Planning:** Goal-directed policy synthesis stripped of extensional literalism.

2.2 Collective Reasoning (The Social Network Domain)

- **Networked Induction and Deduction:** Distributed consensus algorithms and group logic validation.
- **Analogical Transfer over Graphs:** Projecting micro-scale interaction structures onto macro-scale networks.
- **Continuous Group Behavior:** Flocking dynamics, advection, and alignment shifts under localized noise.
- **Intensional Social Planning:** Institutional goal coordination and multi-agent defensive maneuvers.

2.3 Activation and Sovereign Control

The system explicitly addresses the paradigm where an Atemporal Controller (W_2) governs spatiotemporal object execution planes (W_1).

- **Individuals:** Tracking individual salvific and extraction trajectories (e.g., Judith identifying her rescue with the Prodigal Son paradigm; historical trajectories of Moses, David, Paul, and Jesus).
- **Institutions and Collective Behavior:** Designing formal systems for the Courtroom, the Legal System, and the Church (including localized pastoral nodes and the sovereign appointment of disciples).

2.4 Active Domain-Isolated Script Registry

To guarantee rigorous domain isolation and prevent feature cross-talk during execution passes, the repository organizes its core reasoning archetypes into standalone, mutually exclusive script targets. Underscores in script identifiers are properly escaped to protect compilation safety:

1. `query_desert_wandering.lisp`**:** Tracks national-scale generational variance loops and macro-scale orbit trajectories.

2. `query_ecclesial_discernment.lisp`: Convolutes localized Barna micro-splits and macro Pew US Church weather trends to optimize graph edge-weights against noise.

3. `query_food_truck_mentalizing.lisp`: Deduces counterfactual user preferences from agent path deflections around non-convex obstacle bounds.

4. `query_drone_swarm_relays.lisp`: Manages delay-tolerant store-and-forward caching buffers over moving line-of-sight communication meshes.

5. `query_schelling_coordination.lisp`: Evaluates recursive social call-stack reasoning loops to calculate focal point convergence distributions.

6. `query_value_at_risk.lisp`: Tracks Terry's value-at-risk minimization goals under non-linear environmental shocks.

7. `query_matthias_selection.lisp`: Models information-theoretic channel empowerment optimization via the Blahut-Arimoto lot-casting solver.

8. `query_false_belief_task.lisp`: Tokenizes the Pastor Sunday School classroom phone relocation theory of mind matrix.

9. `query_solomon_persuasion.lisp`: Models the 1 Kings 3 trial as a strategic Bayesian persuasion game to collapse guilt probabilities.

10. `query_belief_polarization.lisp`: Evaluates ideological background knowledge variables to track normative divergence thresholds over identical data.

11. `query_temporal_obligation.lisp`: Enforces deontic contract limits and legal invoice payment trajectories over time-bounded steps.

3 Logical Primitives and Narrative Analogies

To express diverse narrative domains, the symbolic layer implements a complete suite of non-classical logics: Classical First-Order Logic (FOL), Edward Zalta's Theory of Abstract Objects, Approximation Logic, Modal/Deontic Logics, Consensus Logic, and Bayesian Probabilistic Programming Languages (PPL). Jon Barwise's local logics and information channels are injected directly into the knowledge bases of the decision forests.

3.1 The Judith-George Situation Isomorphism

The system's capacity for zero-shot analogical reasoning was verified by presenting two distinct situations: a story derived from Barwise concerning Judith requiring extraction from a non-convex mountain crevice, and an structurally identical story regarding George managing a kitchen fire. When mapped through our situation-theoretic descriptors, both narratives successfully converged to the exact same decision tree leaf node, proving that the system classifies core relational patterns rather than literal token names. Similarly, the system successfully parsed Judith's theological transformation by drawing a structural analogy to the Prodigal Son narrative.

3.2 Kinematic Swarm Synthesis and Wavelet Filtration

For collective rescue operations, the Judith scenario was scaled to include 50 autonomous double-integrator mobile drone platforms (UAVs). Reinforcement Learning (RL) agents were deployed to orchestrate optimal search-and-rescue grids. To manage the continuous nature of fluid dynamics and wind field turbulence, Navier-Stokes advection terms and Partial Differential Equations (PDEs) were linked with the controller. Drones utilize mounted sensors running Discrete Wavelet Transforms (DWT) to filter high-frequency noise and extract scale-invariant atmospheric trends as part of the system's active tracking loop.